

Design Specification for Race Walking Data Visualizer

Version 1.0

Prepared By:

Elvis Cao, Ostap Bezrukyy, Jiacong Li, Sotero Casiano, Nico Saputra, Zach Bridgers, Jay Dave

Advisor:

Jeff Salvage

Stakeholder:

Jeff Salvage

1. Document History

Names	Date	Reason	Version
Elvis Cao, Ostap Bezrukyy, Jiacong Li, Sotero Casiano, Nico Saputra, Zach Bridgers, Jay Dave	1/13/2025	Initial Draft	1.0

2. Introduction

2.1 Purpose

This document provides the design specifications for the Race Walking Data Visualizer application. It highlights the design goals of the system and the system overview. The system overview provides context on how the application functions through the visualization of context, architecture, flow charts, and sequence diagrams.

2.2 Overview

The Race Walking Data Visualizer is an application designed to streamline the analysis of race walking events. It provides automated report generation and creates visualizations of athletes' infraction patterns to help analyze athlete performance and infraction metrics.

2.3 Scope

The scope of the application includes a user-friendly interface for accessing and filtering race walking data stored in an SQL database. The application is able to generate detailed reports and visualizations of the race walking data. It also integrates with external data sources through web scraping.

2.4 Design Goals

2.4.1 Modular Architecture:

- The application is structured with a modular architecture, dividing functionality into distinct modules.
- Each module serves specific tasks, fostering code reuse and minimizing redundancy.
- Modular design facilitates easy maintenance, updates, and modifications.
- Clear separation of concerns enhances scalability and adaptability.

2.4.2 Extensibility

- The application offers an extensible framework for seamless integration of new features.
- Developers can create features by extending existing view model classes, reducing development effort.
- This approach ensures consistency in the user interface and facilitates future enhancements.
- Extensibility promotes innovation and accommodates evolving user needs.

2.4.3 Safe Interface

- Measures are in place to prevent data loss due to accidental user actions.

- Features like undo and redo functionalities provide users with a safety net against unintended modifications.
- The interface prioritizes user data integrity and safety.
- User-friendly error handling enhances user experience and confidence in the application.

2.4.4 User-Friendly Interface:

- The application prioritizes user experience with a user-friendly interface.
- Intuitive workflows and navigation paths simplify data access and analysis.
- Clear, concise UI elements enhance usability and reduce user friction.
- User feedback mechanisms promote continuous improvement and optimization.

2.4.5 Scalability and Adaptability:

- The application architecture is designed to scale with evolving user needs.
- Flexible data processing pipelines and modular components support seamless integration of new features.
- Scalability ensures the application can handle increased data volume and user traffic.
- Adaptability enables quick responses to changing requirements and technological advancements.

2.4.6 Reliability and Performance:

- The application prioritizes reliability and performance under varying workload conditions.
- Optimized data retrieval and processing algorithms minimize response times and maximize throughput.
- Rigorous testing and monitoring procedures ensure consistent performance and stability.
- Continuous performance optimization efforts ensure smooth user experience.

2.4.7 Documentation and Support:

- Comprehensive documentation provides users with clear instructions on application usage and features.
- User support resources, including tutorials and online forums, empower users to troubleshoot issues independently.
- Continuous updates and improvements to documentation and support resources enhance user experience over time.
- Dedicated customer support channels offer personalized assistance and guidance to users as needed.

2.5 Definitions

Bent Knee: Angle of an athlete's leg when the foot is in contact with the ground and the leg's position is before it reaches the vertical position. This infraction symbol is denoted by <

Loss of Contact (LOC): Amount of time when both athlete's feet are not in contact with the ground. This infraction symbol is denoted by ~

Python3: A high-level programming language utilized for driving data pulls, data organization, and visualization of graphs within the Race Walking Data Visualizer.

3. System Overview

3.1 Context Diagram

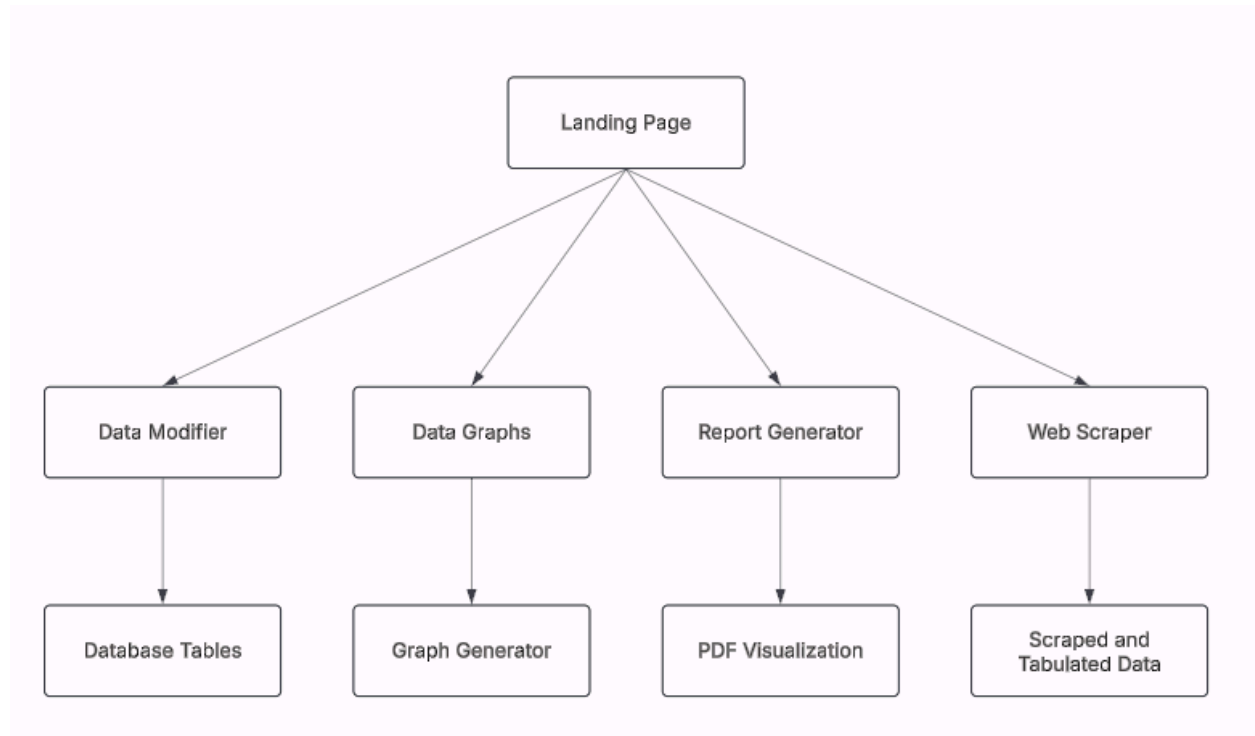


Figure 1 - Displays a high-level representation of the Race Walking Data Visualizer

3.2 Architecture Diagram

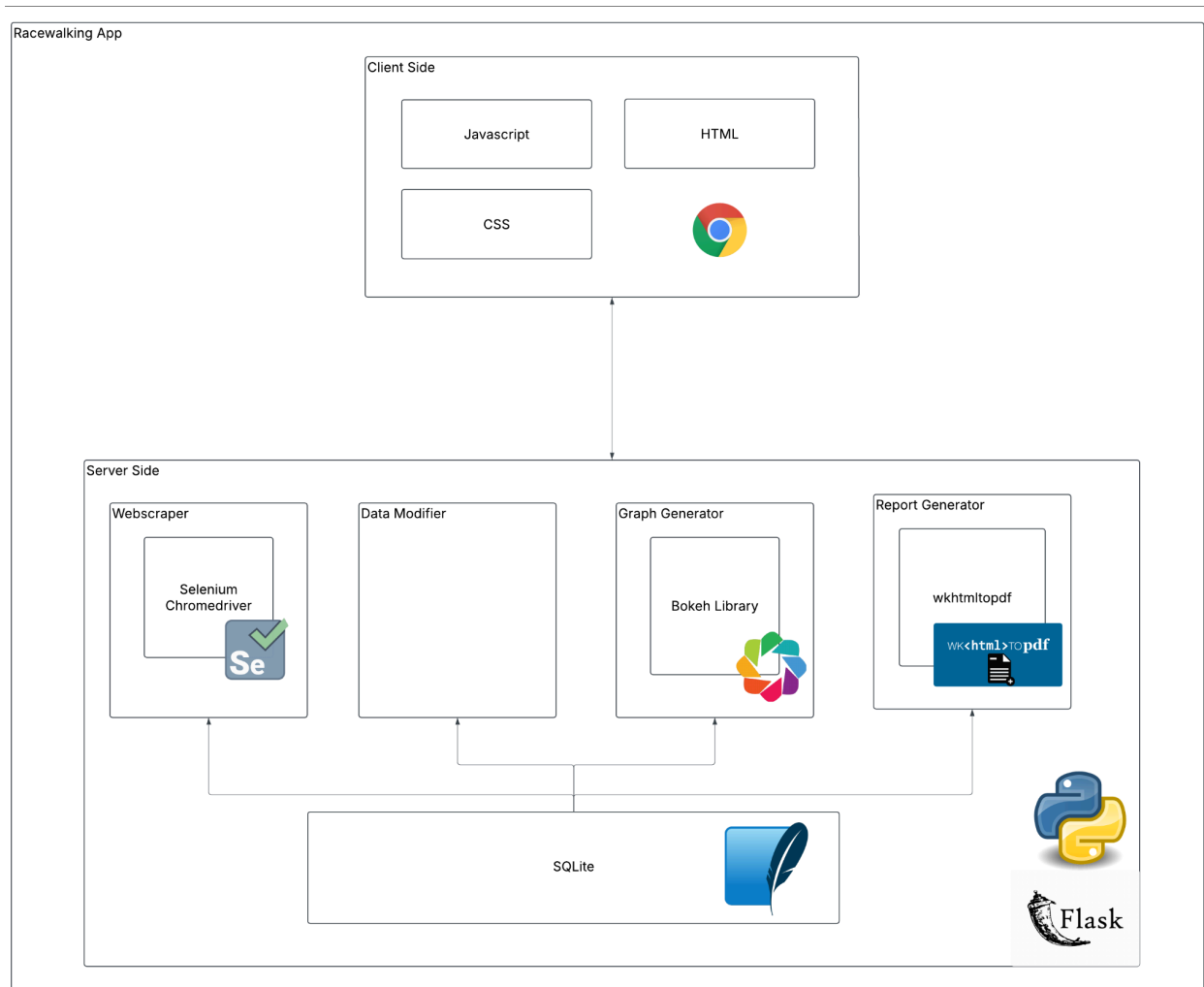


Figure 2 - Displays the relation between the client and server for the Race Walking Data Visualizer

3.3 Flow Charts

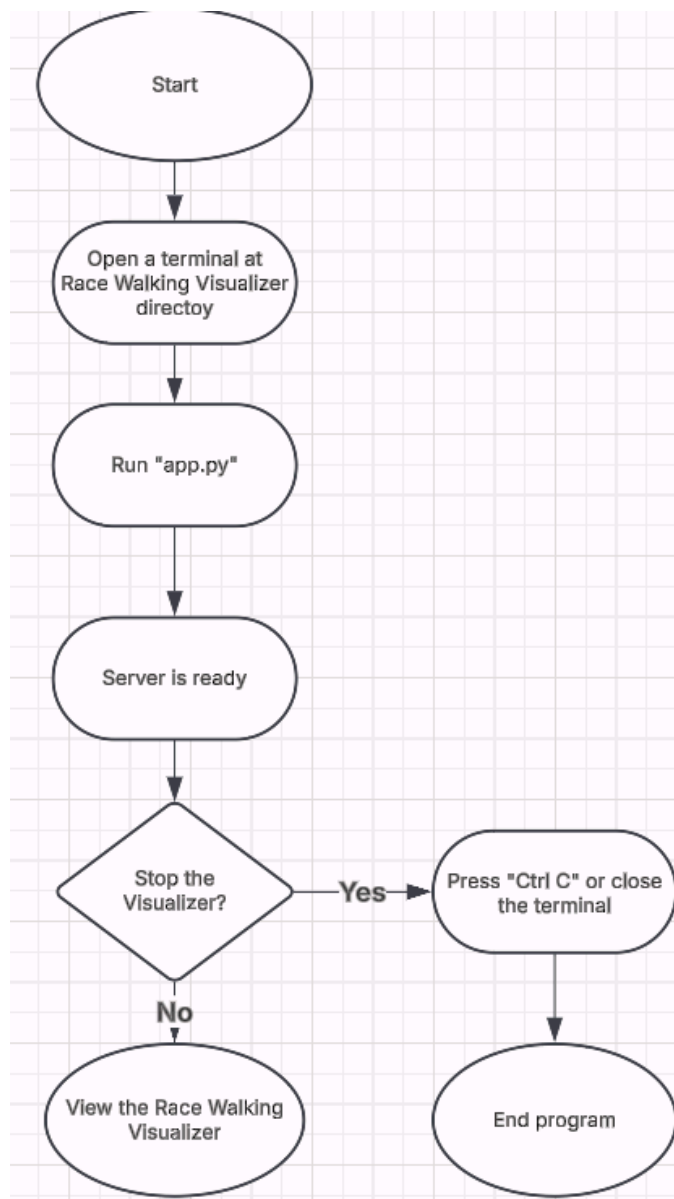


Figure 3 - This flow chart shows the process of launching the app.py server for the application.

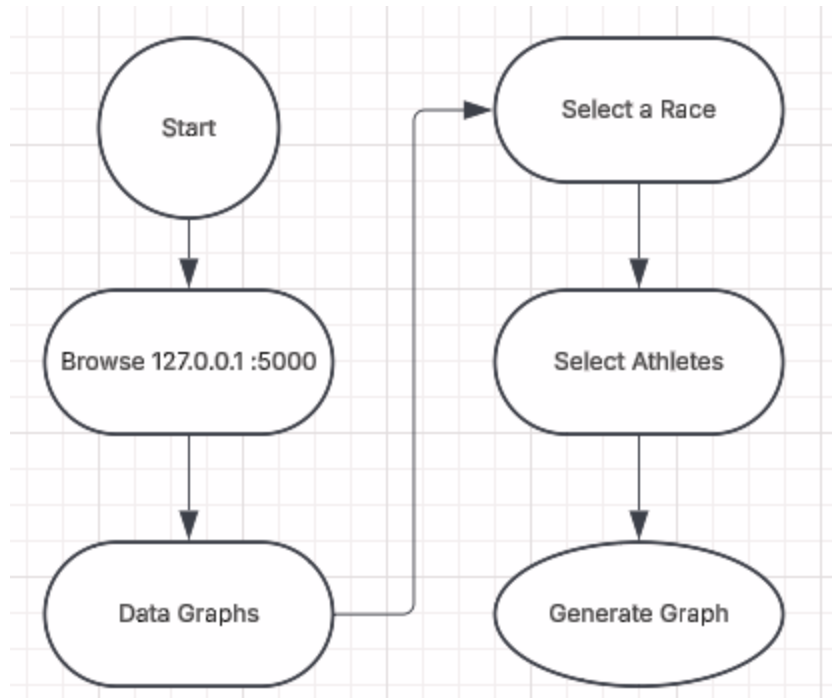


Figure 4 - This flow chart shows the necessary steps taken from launching the Race Walking Data Visualizer application to generating a built-in graph displaying the Loss of Contact vs. Time graph for the athletes selected.

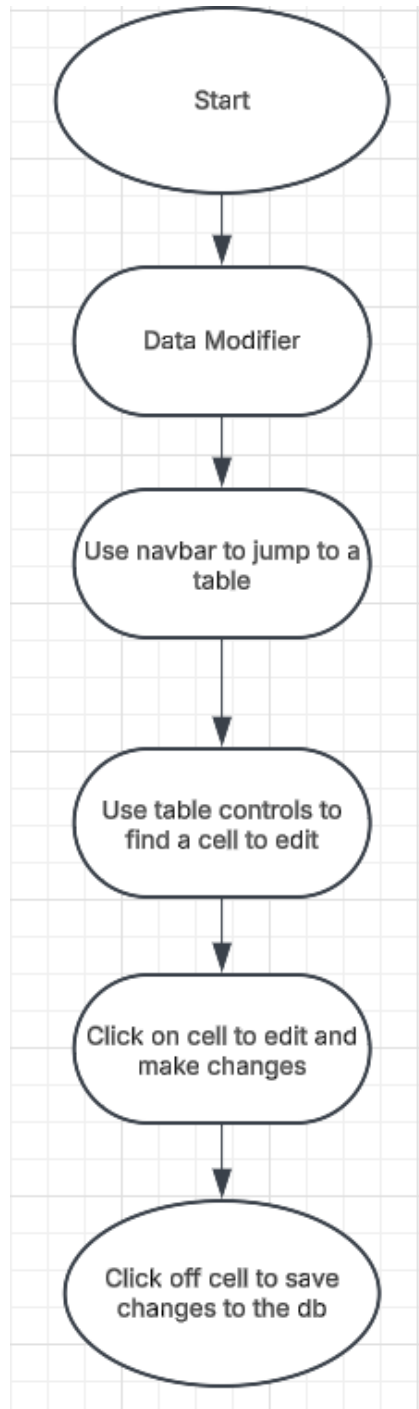


Figure 5 - This flow chart shows the steps taken to navigate the Data Modifier page and edit a table cell from the database.

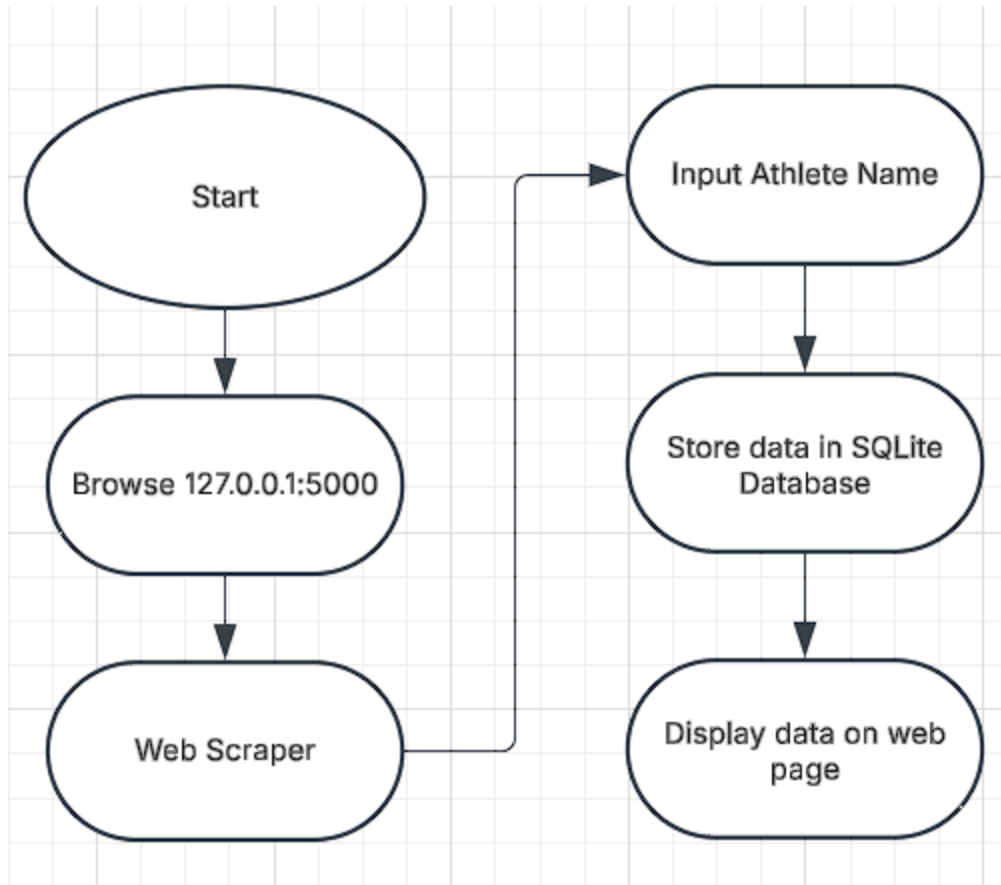


Figure 6 - This flow chart shows the steps taken to search and display an athlete's data from the worldathletics website.

3.4 Sequence Diagrams

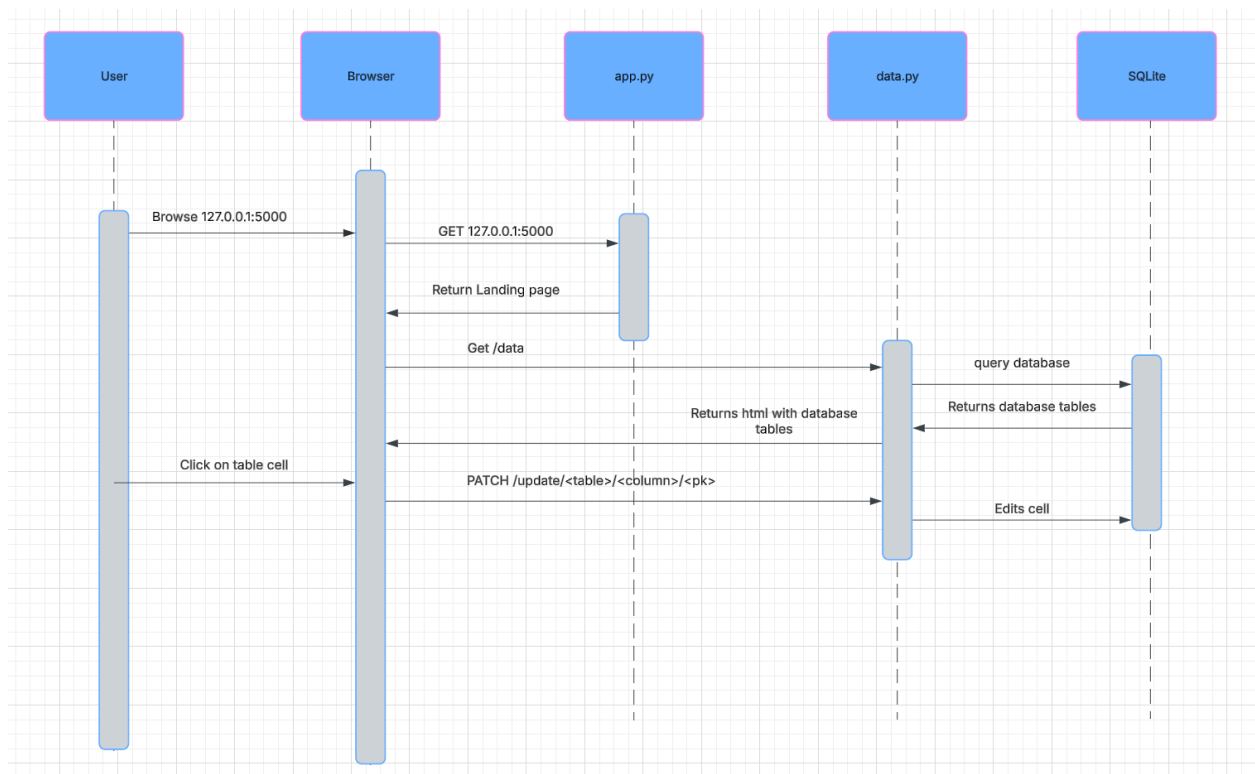


Figure 7 - This sequence diagram shows the steps taken to navigate the Data Modifier page and edit a table cell from the database.

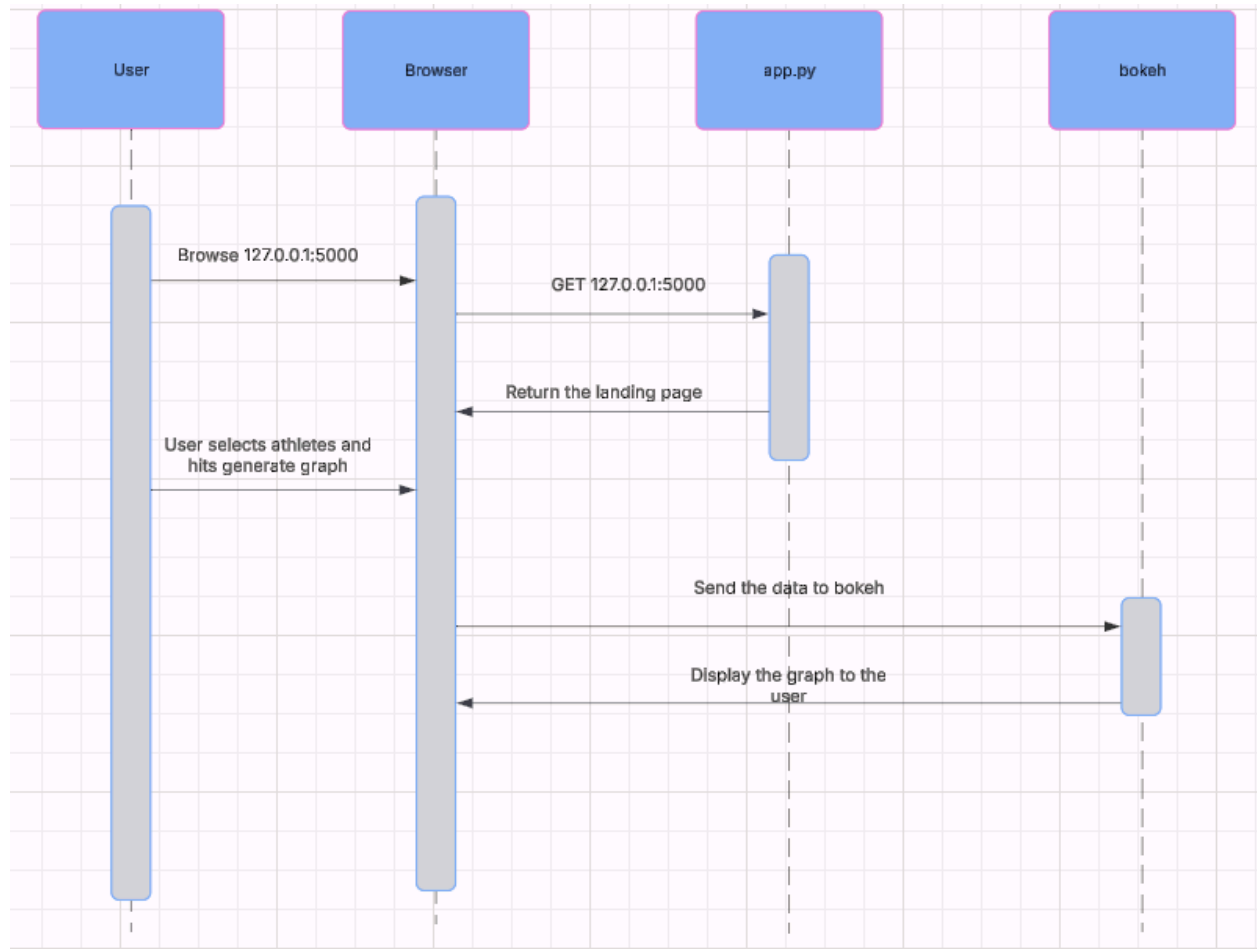


Figure 8 - This sequence diagram highlights the steps taken to generate the graph of the athletes' Loss of Contact (LOC) vs time graph

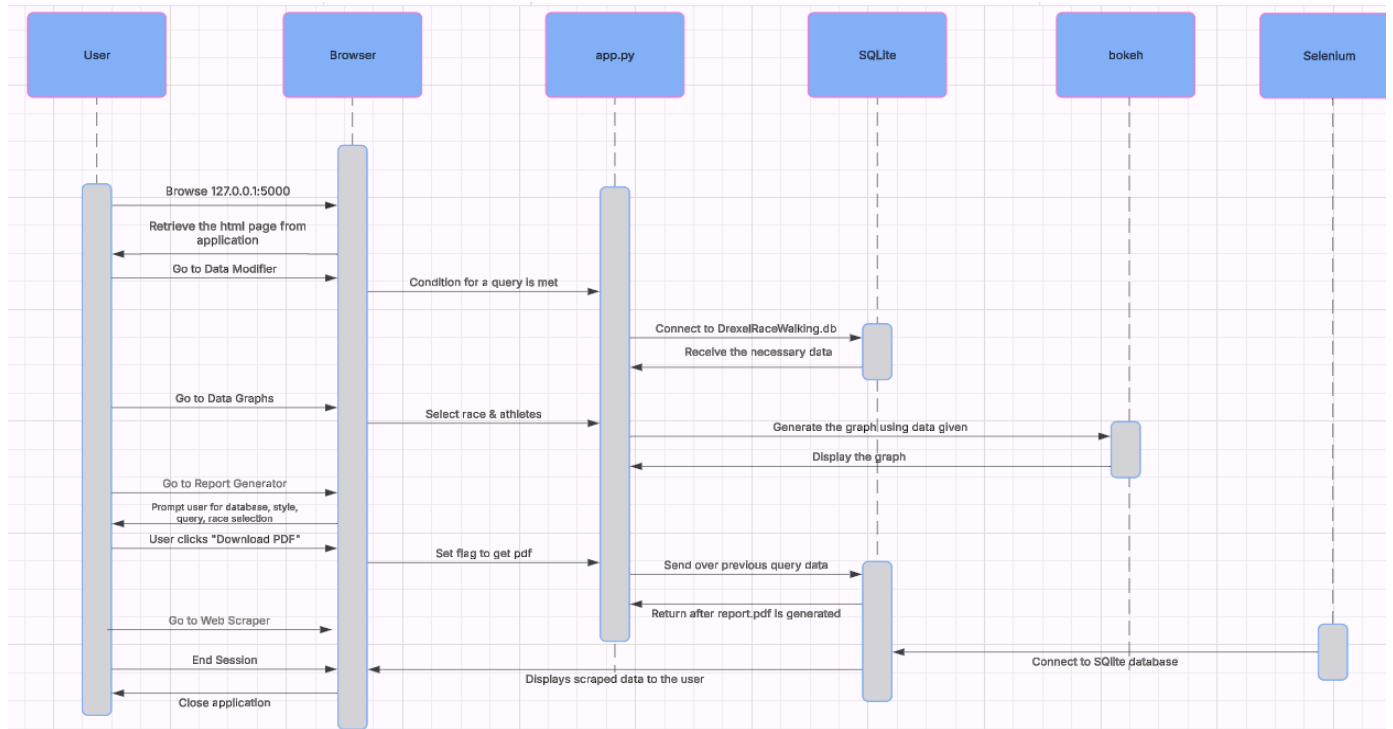


Figure 9 - This sequence diagram displays the user interaction with the application and how it handles user requests

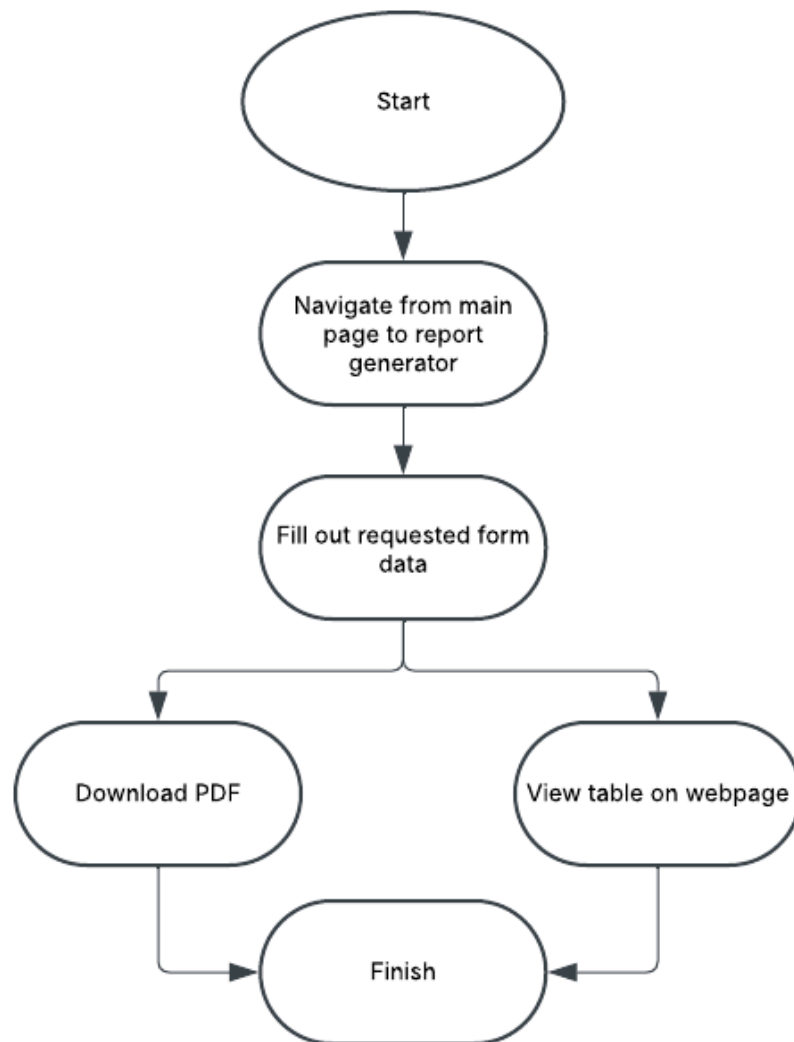


Figure 9 - This flow chart describes the possible means of viewing report data.